DSN 2026 “Find the Language I Am Speaking”: how team godzilla reached first place

Two-page team summary, 23 October 2026. Full write-up: the shared doc and the GitHub repo **Godzilla-lab/dsn2026-lid**.

Where we stand

Our model **D5b scored 0.6222 on the public leaderboard on Wed 21 Oct, first place** (weighted F1 0.6951; the previous leader ray_nacido had 0.6026). We are the only team above zero on Ibibio. Codabench went down at 14:22 UTC on Wednesday and stayed down through the scheduled final phase, so the organisers opened a one-hour window on Thursday. Our final submission there is **D7** (the same recipe retrained with new Ibibio and Nupe voices, expected 0.62–0.66) plus the training notebook.

How the score works

final = weighted F1 × (0.5 + 0.5 × efficiency). Yoruba, Igbo and Hausa count 1/23 each; the other nine languages and “other” count 2/23 each, so **87% of the score sits in the small languages**: one Ibibio clip is worth 14 Yoruba clips. Efficiency mixes size, memory and speed; our 94.6M model scores 0.79, and memory (about 1.7 GB for any HuBERT-type model) is the part we cannot change. Limits: 350M parameters, 1.5 GB, 4 GB memory, 0.5 s per clip, 30 minutes for 6,659 clips. The test mixes clean, noisy and phone-quality audio.

The model (D5b, and D7 on more data)

- **Backbone:** AfriHuBERT (pre-trained on 1,226 African languages), all 12 layers, frozen audio front end, weighted mix of layers. 94.6M parameters, 0.189 GB, 0.006 s per clip on the organisers’ GPU.
- **Balanced sampling:** each step picks a language, then a source, then a speaker, then a clip, so the 3,000-speaker Afrispeech corpus cannot drown the 2–5 speakers we have for Efik, Ibibio or Igala.
- **Augmentation on every crop:** noise floor, speed and pitch, equalisation, room reverb, coloured noise or babble, telephone channel. Noisy accuracy went from 0.21 (our first model) to 0.82.
- **Training:** 8,000 steps of batch 32 on random 3–6 s crops, one-cycle AdamW, about 50 minutes on a Kaggle T4. 12,000 steps gave no gain. 32 English competition clips were relabelled “other”.
- **Weight averaging:** two models from the same head initialisation, different data order, averaged into one model of the same size (+0.02 on held-out speakers).
- **Per-language offsets** written into the classifier bias (no custom code): tuned on speaker-held-out cross-validation, then corrected from D5’s public confusion matrix (Yoruba and Tiv up, Efik and Ibibio down). That correction alone was +0.014.
- **Inference:** the full clip in, one label out. A standard Hugging Face audio-classification model.

The data

Competition data: 24,774 train + 4,352 validation clips (93 h), but Igala 0.5 h, Nupe 0.5 h (no validation clips), Efik 1.4 h, Ibibio 1.8 h, each with 2–6 speakers. We added 62 h of openly licensed audio for exactly those gaps, every set listed in the model cards:

Source	Licence	Used as	Size
Efik speech dataset (Hugging Face)	CC BY 4.0	efi	8.4 h, 2 speakers
FLEURS, NaijaS2ST, nigerian-pidgin, WaxalNLP	CC BY 4.0 / CC BY-SA	hau, ibo, yor; English, Pidgin, Ghanaian languages as “other”	33 h together
VoiceAfrica (NaijaVoices)	CC BY-NC-SA 4.0 + terms	igl, nup, urh (10 speakers each); knc, bcz as “other”	27 h
Wikimedia Nigeria oral histories, Wikitongues, Bloom Library	CC BY-SA 4.0 / CC BY 4.0	ibb, efi, anw, urh, ego, tiv; speech segments only	2 h
Common Voice 27.0 Ibibio (Mozilla Data Collective)	CC0	ibb, 17 speakers (2 dropped)	2.9 h, D7 only
Nupe speech collection (Mozilla Data Collective)	CC BY-NC-SA 4.0	nup, 7 speakers (2 dropped)	2.1 h, D7 only

Every new speaker’s voice was compared with the competition speakers (WavLM x-vectors); matches were dropped so we never train on a test voice. Not used: the three prohibited test corpora, Goloka (no licence), Bible/YouTube audio, and the Efik and Tiv TTS sets (access not granted in time).

Results on the public leaderboard (F1 per language)

	yor	ibo	hau	ego	anw	efi	ibb	idu	igl	nup	tiv	urh	other
D1 (first try, 0.3421)	0.42	0.56	0.60	0.80	0.49	0.00	0.00	0.40	0.03	0.28	0.16	0.86	0.50
D5b (ours, 0.6222)	0.80	0.92	0.93	0.96	0.97	0.34	0.08	0.85	0.35	0.71	0.61	0.99	0.82
ray_nacido (0.6026)	0.80	0.91	0.96	0.96	0.75	0.36	0.00	0.87	0.48	0.59	0.54	0.99	0.86

D5b: accuracy clean 0.81, noisy 0.82, phone 0.78. We lead on Annang, Nupe, Tiv and Ibibio; we trail on Igala and “other”. D7 is not yet scored; on two Ibibio speakers it never trained on it was right 60 times out of 60, where D5b was right 0 times.

What made the difference

• **We optimised the real score.** `src/score.py` reproduces the organisers’ formula, efficiency included; every choice is judged by predicted final score, not accuracy.

• **A local leaderboard.** Speaker-held-out cross-validation, each held-out clip also degraded to noisy and phone, resampled to the test’s language mix. It predicted scores to within about 0.03 and told us what to train before we spent an upload.

• **Speaker-grouped folds.** The given validation split hides the real problem: with 3 Ibibio speakers, a model learns voices, not the language. Our folds keep every speaker on one side; Nupe is measured on held-out VoiceAfrica speakers.

• **Test-matched augmentation** and **new speakers for the weakest languages**: the two largest gains.

• **Reading the organisers’ confusion matrix** and fixing the output layer accordingly: +0.014 from one upload.

• **Proving platform faults instead of guessing.** When every team’s submission failed for a day, byte-identical configs, offline load tests and a 36k-parameter probe model showed it was the runner. We lost two uploads to it, not to our work.

What did not work, and lessons

• Longer training (12,000 steps): no change. Stronger voice and pitch changes to fake new speakers: worse (0.610 vs 0.621). Only real new voices fix a speaker-starved language.

• The local estimate was cautious on D1 (0.31 predicted, 0.34 real) and optimistic on D5 (0.64 vs 0.61): the test’s Tiv and Yoruba speakers are harder than ours. Good for ranking recipes, not for promising a number.

• Keep model repos private until the moment of upload; other teams were downloading ours.

How to score better

• The points left are in Ibibio (0.08), Efik (0.34), Igala (0.35) and Tiv (0.61): 35% of the score, averaging 0.35. Everything else is above 0.80.

• Next: the Efik-TTS and Tiv-TTS sets from the Mozilla Data Collective (terms must be accepted on the site; the pipeline then takes 20 minutes plus a 50-minute retrain); more Igala speakers from a different recording style; a three-way weight average (+0.005); distillation from D7 (+0.01 to +0.02).

• Each new confusion matrix allows one more small, held-out-checked correction of the output offsets.

Where everything lives, and quick answers

GitHub **Godzilla-lab/dsn2026-lid**: `STATUS.md` (state), `submissions/` (every zip, `LOG.md`, the solution notebook), `src/` (training, scoring, packaging, data intake), `results/` (analyses), `research/` (data manifests). Hugging Face: `godzilla1215/dsn2026-lid-d7` (final, revision 3d5425c9), `-d5b` (scored 0.6222, 4ea16f1e), `-d5c`; all public. Kaggle account adewumiadenike holds the private training datasets.

• **Why AfriHuBERT?** It knows our languages; Whisper only knows Yoruba and Hausa among them. A smaller model would gain at most 2% from efficiency and lose far more accuracy.

• **Did we touch the test data?** No. We only read the per-language scores and confusion matrix the organisers show every team.

• **Is averaging two models allowed?** Yes: the result is one ordinary model of the same size; the rules restrict the format, not how the weights were obtained.

• **Why is Ibibio so low for everyone?** Three training speakers, and Ibibio is close to Efik and Annang. D7 adds 15 new voices.

• **How long did it take?** Two days: research Tuesday morning, first upload Tuesday 13:22 UTC, first place Wednesday 11:17 UTC.